



Kaleidospheres

Design, Manufacturing, and Future Improvement

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Executive Summary

This report presents the results of the Kaleidosphere project, conducted by Ethan Goldman, Tyler Pisinski, and May Wildgrube under the supervision of Professor James Intriligator. The project aimed to transform Professor Intriligator’s conceptual “Moiré Spheres” into a tangible and manufacturable toy: the “Kaleidosphere”. These innovative spheres utilize dynamic moiré patterns to create captivating optical effects, offering educational value by enhancing spatial reasoning and visual perception. The Kaleidosphere is designed to appeal to young children and their families, with potential for retail in academic and science-themed stores.

To achieve this goal, the team developed and refined prototypes through iterative design and testing. Early proof-of-concept models were created using commercially available materials, employing a custom underwater assembly process to encapsulate a weighted inner sphere within an outer sphere. These prototypes successfully demonstrated the feasibility of generating dynamic moiré patterns through rotation.

For manufacturability, the team explored scalable production methods, ultimately focusing on injection molding for its efficiency and precision. Initial attempts to mold the spheres in-house revealed limitations in available equipment, particularly the inability to reach the temperatures required to mold polycarbonate. While COMSOL was not directly utilized during the project to optimize the molding process, it presents a valuable opportunity for future development. This computational modeling software could be employed to design more effective molds, enabling precise control over parameters such as flow dynamics, cooling rates, and material behavior during the injection process.

Additionally, the team developed a reliable hydro-dipping process to apply high-quality, visually engaging patterns to the spheres. This method allows for consistent and scalable application of the moiré patterns, a critical feature of the toy’s functionality and appeal.

The final design effectively balances innovation, user engagement, and manufacturability. It provides an accessible, visually stimulating, and educational experience for children while demonstrating strong potential for commercialization in niche toy markets.

Introduction

Problem Statement

Moiré patterns are captivating optical effects created when overlapping patterns of lines or grids interact with each other at varying angles. Professor James Intriligator at Tufts University envisions leveraging these patterns to design toys that not only captivate visually but also enhance children's spatial reasoning and visual perception. The proposed toy concept involves a spherical, hand-held design that transfers the traditional two-dimensional design of moiré patterns onto a three-dimensional toy. The moiré patterns should be dynamically created when the toy is rolled or rotated.

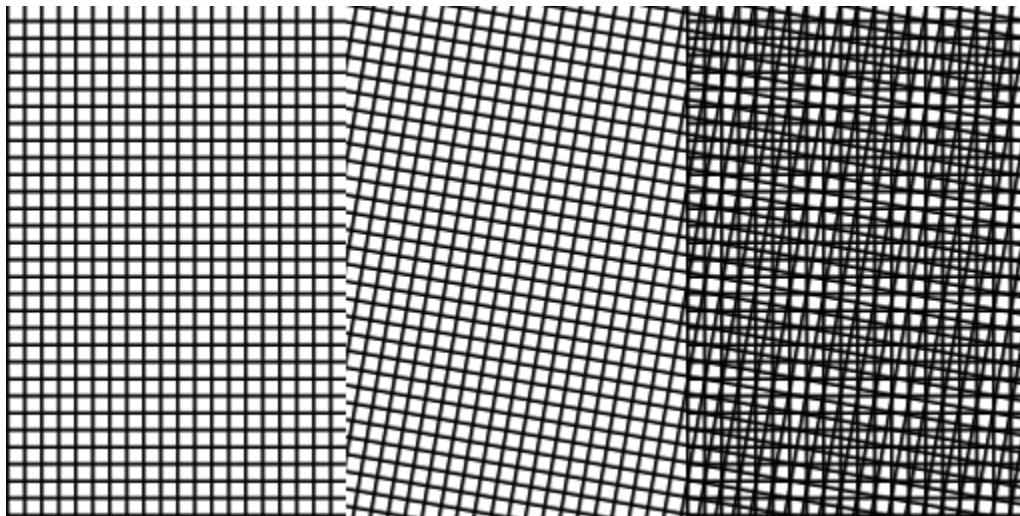


Fig. 1. Moiré pattern (right) created by superimposing two grids (left and middle) [1].

Motivation

Visual perception and spatial reasoning are fundamental cognitive skills that shape how children understand and interact with the world around them. According to *Behind the Behavior Psychology*, “Visual Spatial Functioning ... can greatly impact how [a] child demonstrates their knowledge in subjects like math, geometry, science, and reading, as well as their ability to engage with peers, perform gross motor tasks, and safely interact within their environment” [2]. Despite the importance of these skills, tools that effectively teach these concepts in an engaging

and interactive manner are limited. Yang et al. found that, “diverse training strategies or programs including hands-on exploration, visual prompts, and gestural spatial training significantly foster young children's spatial skills.” [3]. Moiré patterns, with their striking and dynamic visual effects, offer a unique opportunity to bridge this gap by providing an intuitive and captivating way to explore these principles.

Professor James Intrilligator identified the potential for toys utilizing moiré patterns to serve as both education tools and sources of entertainment. By stimulating curiosity and encouraging hands-on interaction, such toys could deepen children’s understanding of visual phenomena while fostering cognitive development. Furthermore, creating a toy that is spherical and hand-held introduces a tactile and immersive experience that complements the visual appeal of the moiré patterns.

The goal of making these toys widely available adds another level of importance to this project. By designing the toy for manufacturability at scale and ensuring low production costs, it becomes possible to make these educational tools accessible to a broad audience. This accessibility aligns with the broader goal of democratizing educational resources and inspiring a new generation to appreciate the intersection of science, design, and play.

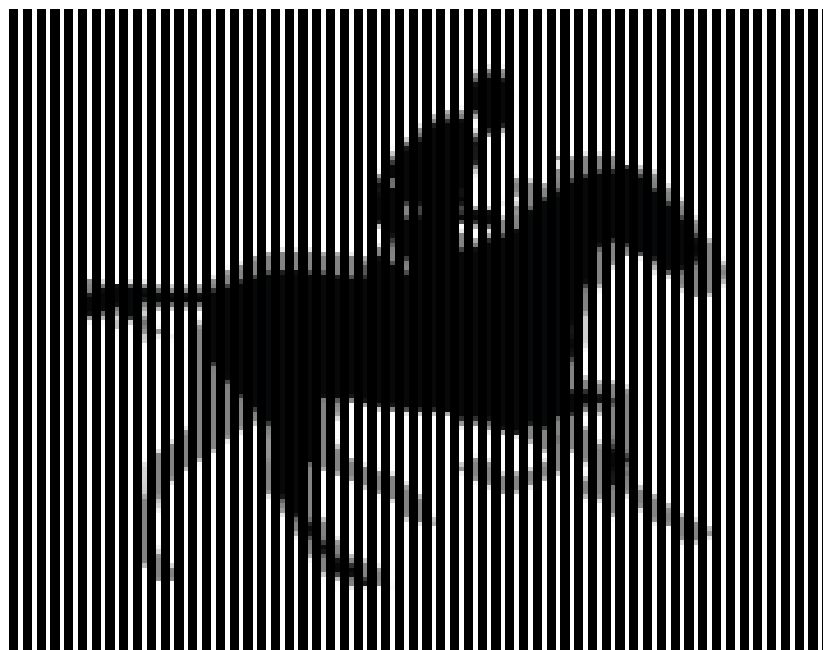


Fig. 2. Moiré patterns can create engaging and fun patterns such as a person riding a horse which actually appears to move due to the optical illusion [4].

Design Solution

User Needs

The design of the moiré pattern toy is driven by a set of user needs which has been identified through research, client input, and iterative development. These needs ensure the product is visually engaging, practical, and suitable for large-scale production. Below are the key user needs we identified:

1. Visually Stimulating and Engaging

The toy must produce captivating moiré patterns that dynamically respond to user interactions. This visual stimulation is critical to maintaining interest and supporting the toy's educational purpose.

2. Affordable

The toy must be cost-effective to produce and purchase, ensuring accessibility to a wide audience, including families, schools, and retailers.

3. Safe for Repeated Play

As a child-focused product, the toy must meet relevant safety standards to ensure it is safe for frequent use. Its spherical design inherently eliminates risks from sharp edges, but the materials and construction must also be free from small, detachable parts or hazardous materials.

4. Durable

The toy must be robust enough to withstand frequent handling, rotational movements, and accidental drops, ensuring longevity and reliability.

5. Lightweight

To facilitate ease of use and portability, the toy must be lightweight and comfortable for children to handle and manipulate during play.

6. Unique Use of Moiré Patterns

The toy must creatively and uniquely incorporate moiré patterns in its design, ensuring that it stands out as an innovative educational and entertainment tool.

7. Marketable

The toy's design should appeal to both children and adults, including educators and parents, making it an attractive product for both retail and educational markets.

8. Manufacturable at Scale

The toy must be designed with scalable production methods in mind, ensuring that it can be manufactured efficiently and at low cost without compromising quality or performance.

Engineering Requirements

The engineering requirements for the moiré pattern toy are designed to ensure alignment with user needs, manufacturability, and relevant standards. These requirements guide the design and provide measurable criteria to validate its success. Below is a description of each requirement and its rationale. A full matrix relating user needs to engineering requirements is provided in Appendix II.

1. Effective Moiré Effect

- a. **Target:** At least 9 out of 10 users must perceive a clear moiré effect during interaction with the toy.
- b. **Rationale:** Ensures the toy fulfills its primary visual engagement purpose.

2. Weight

- a. **Target:** The toy must weigh no more than 0.5lbs (0.23kg).
- b. **Rationale:** A lighter design ensures ease of handling, particularly for children.

3. Manufacturing Cost

- a. **Target:** Total manufacturing cost per toy must not exceed \$3.
- b. **Rationale:** Keeps the toy affordable while maintaining profitability at scale.

4. Dimensions

- a. **Target:** Outer sphere diameter must be between 2 and 4 inches.
- b. **Rationale:** Balances ergonomic design with visual effectiveness of the moiré patterns.

5. Durability

- a. **Target:** The toy must withstand a load of at least 75 N without permanent deformation or failure.
- b. **Rationale:** Ensures durability for repeated use and accidental drops.

6. Manufacturing Efficiency

- a. **Target:** Fabrication time per unit must not exceed 1 hour.
- b. **Rationale:** Supports scalability and minimizes production costs.

7. Operational Lifetime

- a. **Target:** The toy must retain full functionality for at least 2 years of use.
- b. **Rationale:** Provides value to consumers and reduces waste.

8. Interactive Moiré Patterns

- a. **Target:** Patterns must be generated through dynamic interaction between the rotating outer sphere and stationary inner sphere.
- b. **Rationale:** Ensures the toy demonstrates the core concept of moiré effects effectively.

9. Material Safety and Compatibility

- a. **Target:** Materials must be non-toxic, comply with relevant safety standards, and be suitable for mass production techniques.
- b. **Rationale:** Ensures safety and manufacturability at scale.

Compliance with Relevant Standards

To ensure the moiré pattern toy meets safety and quality expectations, we have identified and incorporated the following key standards into the design and manufacturing process.

ASTM F963-23: Standard Consumer Safety Specification For Toy Safety [5]

ASTM F963 is a widely recognized standard for toy safety, applicable to toys designed for children under the age of 14. It addresses potential hazards associated with materials, design, and use, ensuring that products meet safe regulations.

Application to the Project

This standard applies to several aspects of the moiré pattern toy, including:

- **Material Safety:** Ensures that all components are non-toxic and free from hazardous chemicals.
- **Structural Integrity:** Requires the toy to withstand normal play conditions, including drops and handling, without breaking into hazardous pieces or creating sharp edges.

- Choking Hazards: Prohibits the inclusion of small, detachable parts that could pose choking risks to children.
- Surface Coating: Regulates the use of paints, inks, or finishes to ensure they do not contain lead or other toxic substances.

Ensuring Compliance

In order to ensure that the final iteration of the moiré pattern toy meets the requirements of ASTM F963-23, the final product should be produced with the following specifications:

1. Material Selection: All materials will be sourced from certified suppliers, ensuring they meet non-toxic and child-safe criteria.
2. Product Testing: The product will undergo testing for impact resistance, structural integrity, and material safety using ASTM protocols.
3. Certification: The final product will be tested by a third-party lab for ASTM F963 certification before mass production.

Importance of Standards

Adhering to safety standards is essential for ensuring the moiré pattern toy is both marketable and safe for repeated play. Compliance not only protects users but also builds consumer trust and meets legal requirements for toy distribution. By following these standards, the toy will be positioned as a reliable, child-safe product suitable for large-scale production.

Competitive benchmarking

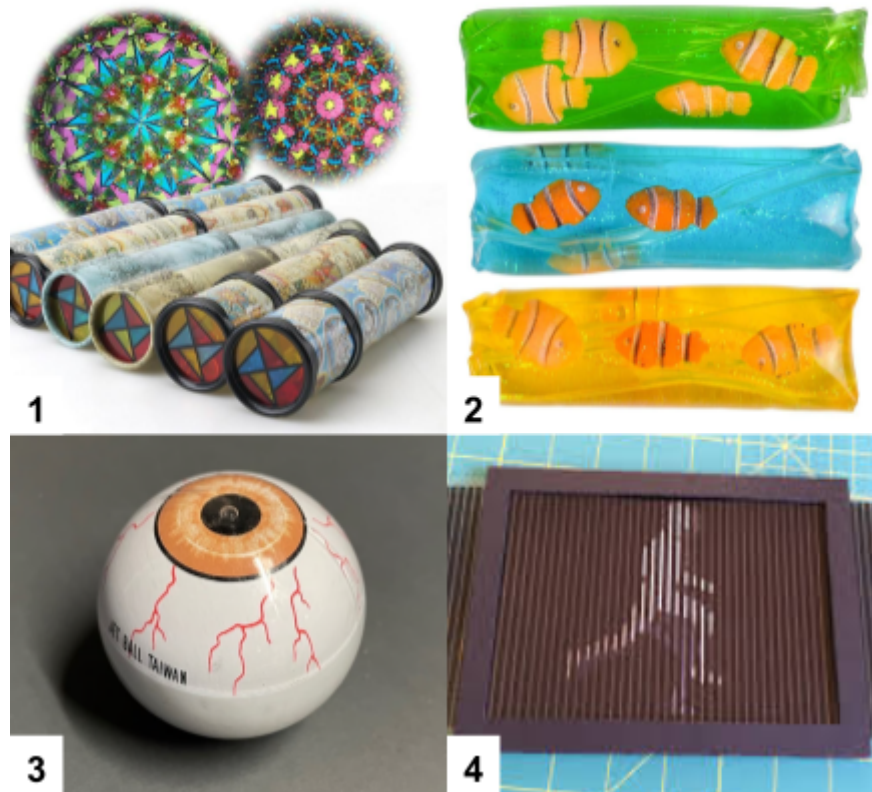


Fig. 3. Toys similar to that of the Kaleidosphere. (1) Kaleidoscope Toys [6], (2) Liquid-filled Toys [7], (3) Jet/Glide Balls [8], (4) DIY Moiré Pattern Kits [9].

To evaluate the design of the moiré pattern toy, we analyzed four different solutions in the toy market that involve visual effects and interactive features. This benchmarking highlights the strengths and limitations of these products in meeting white identified user needs and engineering requirements, while also demonstrating the unique value of our design.

1. Kaleidoscope Toys

- a. **Description:** Kaleidoscopes are hand-held toys that create shifting patterns through mirrored surfaces and colored beads or lenses.
- b. **Strengths:** Visually stimulating and widely appealing, durable and inexpensive to manufacture.
- c. **Weaknesses:** While patterns can be user-modified, they are not dynamically created through interactions between separate elements. The focus is on visual appeal rather than an educational exploration of optical phenomena.

- d. **Comparison:** The moiré pattern toy builds on the interactive appeal of kaleidoscopes by allowing users to dynamically generate patterns through spherical rotation, providing both visual engagement and an opportunity to learn about the science of moiré effects.

2. Liquid-Filled Toys

- a. **Description:** Toys featuring dynamic liquid movement, such as squishy tubes or timers where colored liquid flows to the bottom.
- b. **Strengths:** Highly engaging due to the tactile and visual appeal; encourages sensory exploration; lightweight and portable.
- c. **Weaknesses:** Limited durability due to leakage risks; lacks educational value tied to spatial reasoning or moiré patterns.
- d. **Comparison:** While visually appealing, liquid-filled toys do not offer the interactive or educational focus of the moiré pattern toy, while fostering hands-on learning about optical phenomena.

3. Jet/Glide Balls

- a. **Description:** Ball-shaped toys that have a sphere contained within another sphere. When the toy is rolled across the floor, the inner sphere maintains a constant orientation making whatever pattern is on the inner sphere appear to be gliding across the floor.
- b. **Strengths:** Engaging for tactile play; inexpensive to produce and purchase. Has a similar morphology to that of the moiré pattern toy to be produced.
- c. **Weaknesses:** No moiré patterns present in the product; lacks educational focus and often does not leverage dynamic visual effects in a meaningful way.
- d. **Comparison:** While jet/glide ball toys offer tactile engagement and visual appeal, the moiré pattern toy emphasizes dynamic interaction between patterns, making it both visually engaging and educationally valuable.

4. DIY Moiré Pattern Kits

- a. **Description:** Educational kits that allow users to assemble and interact with moiré patterns using pre-printed sheets and templates.
- b. **Strengths:** High educational value; introduces users to the science of moiré patterns; cost-effective.

- c. **Weaknesses:** Lacks durability as the components are often paper-based; limited engagement due to static patterns.
- d. **Comparison:** The moiré pattern toy provides a more durable and interactive experience, offering the ability to generate patterns dynamically with spherical components while still teaching the underlying principles of moiré patterns.

		JET BALL	KALEIDOSCOPE TOYS	LIQUID-FILLED TOYS	DIY MOIRÉ PATTERN TOYS
NEEDS	Engaging	✓	✓	✓	
	Safe	✓	✓		✓
	Durable	✓	✓		
	Lightweight	✓	✓	✓	✓
	Unique use of moiré patterns				✓

Fig. 4. Comparison of existing solutions to key user needs. Solution images are from Fig. 3.

Concept generation, selection, and iteration

Two primary design concepts were explored to develop the moiré-pattern toy, each representing a different approach to balancing user needs, functionality, and production feasibility.

1. Christmas-Ornament Design, Taped Patterns

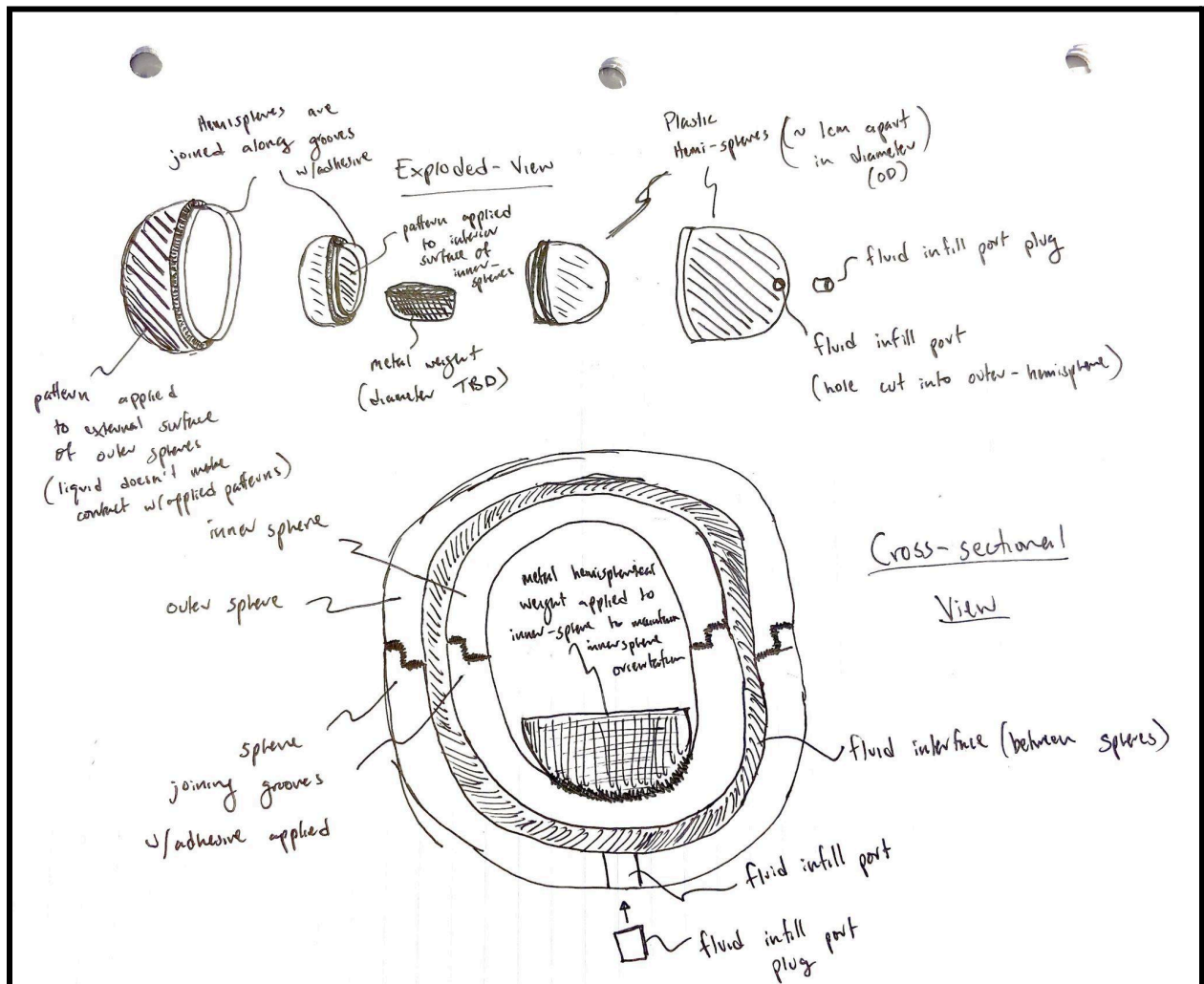


Fig. 5. Christmas-Ornament Design, Taped Patterns concept sketch.

- **Description:**

- Clear acrylic Christmas ornaments were used to form the inner and outer spheres. The inner sphere was painted white to enhance contrast and weighted with metal ball bearings secured with super glue. During assembly, the gap between the spheres was filled with water by

submerging the components underwater and assembling the outer sphere around the inner sphere. The outer sphere was sealed using a press-fit and super glue.

- Moiré patterns were created using black vinyl tape applied to the exterior of the inner sphere and the interior of the outer sphere.
- **Outcomes:**
 - This design served as an early-stage prototype for testing the visual dynamics and rotational functionality of the moiré patterns.
 - Lightweight materials and simple assembly made it a cost-effective prototyping method.
 - Limitations included durability issues, leakage issues, and manual assembly processes, which were not scalable for large-scale production.

2. Injection-Molded Design, Hydro-Dipped Patterns

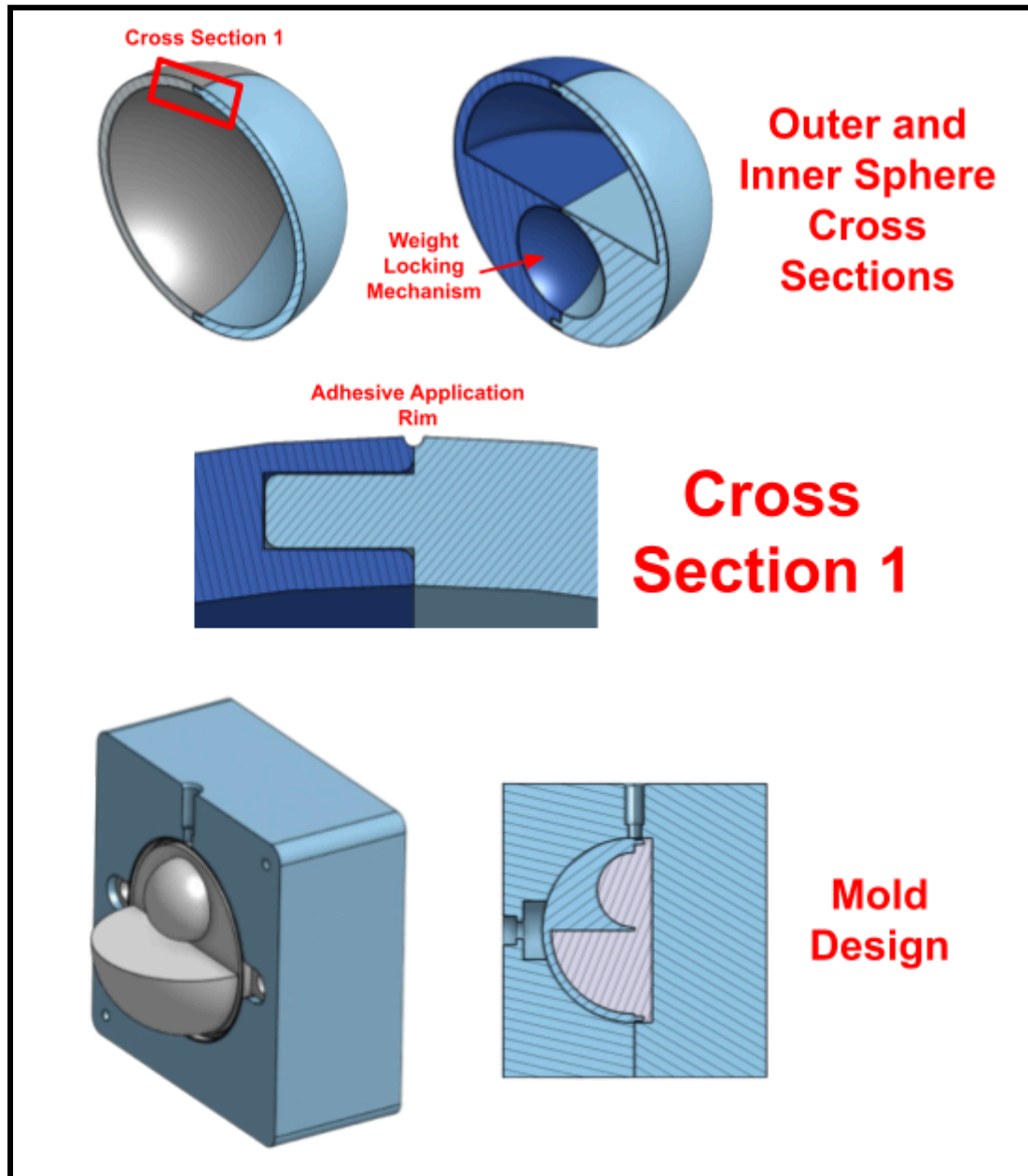


Fig. 6. Injection Molded Design, Hydro-dipped Patterns concept sketch.

- **Description:**

- Injection-molded polypropylene was used to create the spheres. Custom molds ensured a precise gap between the inner and outer spheres. A built-in weight was integrated into the inner sphere using a locking

mechanism rather than adhesives. The two hemispheres of the outer sphere were joined with a low-viscosity adhesive applied uniformly around a rim designed for this purpose.

- Hydro-dipping techniques were employed to apply high-quality, consistent moiré patterns to the spheres.
- **Outcomes:**
 - This design addressed durability concerns with robust materials and a secure assembly method.
 - Manufacturability at scale was demonstrated due to the use of custom molds and streamlined production methods.
 - Polypropylene reduced the transparency, strength, and durability of the spheres compared to the ideal material polycarbonate. These limitations are temporary, however, resulting from equipment constraints during prototyping.

Exploration of the Design Space

Comprehensive exploration of the design space was achieved through ideation techniques, including brainstorming sessions, rapid prototyping, and morphological analysis . A morphology chart was created to investigate variations in toy design and pattern application techniques. The full morphology chart can be viewed in Appendix I.

Considering User Needs in Concept Selection

User needs were central to evaluating and selecting the final design. Each concept was assessed for its ability to meet these needs:

- **Visual Engagement:** Both designs produced dynamic moiré patterns, but the hydro-dipping method in the Injection-Molded Design resulted in more consistent and polished visuals.
- **Durability and Safety:** The Christmas-Ornament Design lacked robustness for frequent use and faced challenges in water retention. The Injection-Molded Design used stronger materials and an improved hemisphere-joining interface, ensuring safety and longevity.

- **Scalability:** While the Christmas-Ornament Design was effective for small-scale prototyping, it relied on manual assembly, making it unsuitable for mass production. The Injection-Molded Design resolved this issue with automated processes and custom molds.
- **User Experience:** Both designs prioritized lightweight construction. However, the Injection-Molded Design offered a professional finish and ease of use, making it more appealing to end users.

A decision matrix was used to compare the two concepts systematically, with criteria weighted according to user needs:

Criteria	Weight	Christmas-Ornament Design	Injection-Molded Design
Visual Appeal	0.15	8	9
Durability	0.15	6	9
Manufacturability	0.2	5	9
Cost	0.25	5	8
User Safety	0.25	5	8
Weighted Total	1.0	5.60	8.50

Fig. 7. Decision matrix evaluating the success of the two prototypes in meeting user needs.

Justification of the Final Selected Concept

The **Injection-Molded Design** was chosen as the final concept because it best met the user needs and project goals (see Fig. 7). Key advantages included:

1. **Visual Appeal:** Hydro-dipping ensured captivating and consistent moiré patterns.
2. **Durability:** Robust materials addressed concerns about wear and tear.
3. **Scalability:** Custom molds and industrial production methods allowed for efficient and cost-effective manufacturing.
4. **Refinement Through Iteration:** Insights from the Christmas-Ornament Design were incorporated to improve functionality and the manufacturing process.

How Iterative Design Improved Alignment with User Needs

The iterative design process allowed for continuous refinement and alignment with user needs.

1. **Early Prototyping:** The Christmas-Ornament Design provided a low-cost method for validating visual and functional requirements.
2. **Material and Manufacturing Enhancements:** Transitioning to injection molding and hydro-dipping improved scalability and durability while maintaining the toy's core functionality.
3. **Data-Driven Decision-Making:** The decision matrix highlighted trade-offs and ensured the optimal design was selected.

Through these iterations a manufacturable prototype was created that balances user needs with production feasibility.

Engineering calculations and analysis

Engineering analysis was conducted to investigate the failure of the injection molding process, focusing on the underlying causes and opportunities for improvement. A detailed review of polymer flows was performed, examining how material properties and flow behavior contributed to the observed issues. The failure was attributed to irregularities in the polymer flow during the molding process. Polymer flow is a non-newtonian flow, which means that viscosity can't be treated as a constant and elastic effects can't be ignored¹. These effects make Newton's law of viscosity invalid, and another equation must be used instead. In this case we transform equation 1, Newton's law of viscosity for incompressible flow, and replace the constant viscosity μ with the dynamic viscosity parameter η , where η is a function of the magnitude of the rate of strain tensor γ defined in equation 1.

$$\boldsymbol{\tau} = -\mu(\nabla \mathbf{v} + (\nabla \mathbf{v})^\dagger) = -\mu\boldsymbol{\gamma} \quad \text{eqn 1}$$

$$\boldsymbol{\tau} = -\eta\boldsymbol{\gamma} \quad \text{eqn 2}$$

$$\text{where } \frac{\eta - \eta_\infty}{\eta_0 - \eta_\infty} = [1 + (\lambda\gamma)^2]^{(n-1)/2} \quad \text{eqn 3}$$

¹ From ME150: Thermofluids Transport

Equation 3 is the Carreau equation, which we can use to define a dynamic viscosity for our material, polypropylene. For polypropylene, these values correspond to $\eta_0 = 3199 \text{ Pa}\cdot\text{s}$, $\lambda = 0.508 \text{ s}$, and $n = .736$ for the specific PP from ISPLEN PP-040G1E Repsol YPF [10]. From these equations, it is apparent that since the stress tensor is proportional to the rate of strain tensor, as strain increases, viscosity increases according to equation 3. The rate of strain is also determined by the change in viscosity in all three directions, which is constantly changing for a flow that is both transient and going around a sphere. While we did not have the proper setup directly measure enough quantities to calculate the stress tensor or rate of strain tensors in our set up (this would require complex imaging), we can determine that our shape is overall worse than the shapes typically used in the machine we used, as the sphere itself creates more viscosity.

In an ideal world, a full CFD simulation would have been run. However, limitations in computational resources prevented a full simulation, as this would require a transient simulation of a viscoelastic flow, something that is extremely computationally expensive. Furthermore, little can be done to simplify it and get applicable values for heat transfer, inlet pressures required, and fill speeds, since there is no steady state for any of these issues. Due to these complications, no CFD study was conducted; however, it is highly suggested that one be run in a future study with more resources to get a proper understanding of the molding process

Final Design

The final design consists of a **polycarbonate inner sphere** suspended in water within a **polycarbonate outer sphere**. This innovative design uses hydro-dipping to create intricate patterns on both the inner and outer spheres, producing visually captivating moiré effects when the toy is rotated or rolled. The durable polycarbonate material ensures clarity and strength, while the water interface minimizes friction and enhances the smooth rotational movement of the inner sphere.

Design Features and Components

- **Inner Sphere:**
 - **Material:** Polycarbonate, for its transparency and durability.

- **Weight Mechanism:** A built-in weight ensures the inner sphere remains upright, creating dynamic interactions between the patterns on the inner and outer spheres. The weight is securely housed in a locking mechanism integrated into the sphere.
- **Pattern:** Hydro-dipped black patterns on the exterior surface of the inner sphere interact with the patterns on the outer sphere to generate the moiré effect.
- **Outer Sphere:**
 - **Material:** Polycarbonate, for its transparency and durability.
 - **Pattern:** Hydro-dipped black patterns on the interior surface of the outer sphere complement the patterns on the inner sphere.
- **Water Interface:**
 - The space between the inner and outer spheres is filled with water to reduce friction and facilitate smooth movement.
- **Assembly Features:**
 - The hemispheres are joined using a precise press-fit and sealed with low-viscosity adhesive to ensure a watertight seal.
 - A small rim around the joining edges of the sphere hemispheres simplifies the application of adhesive, ensuring uniform sealing.

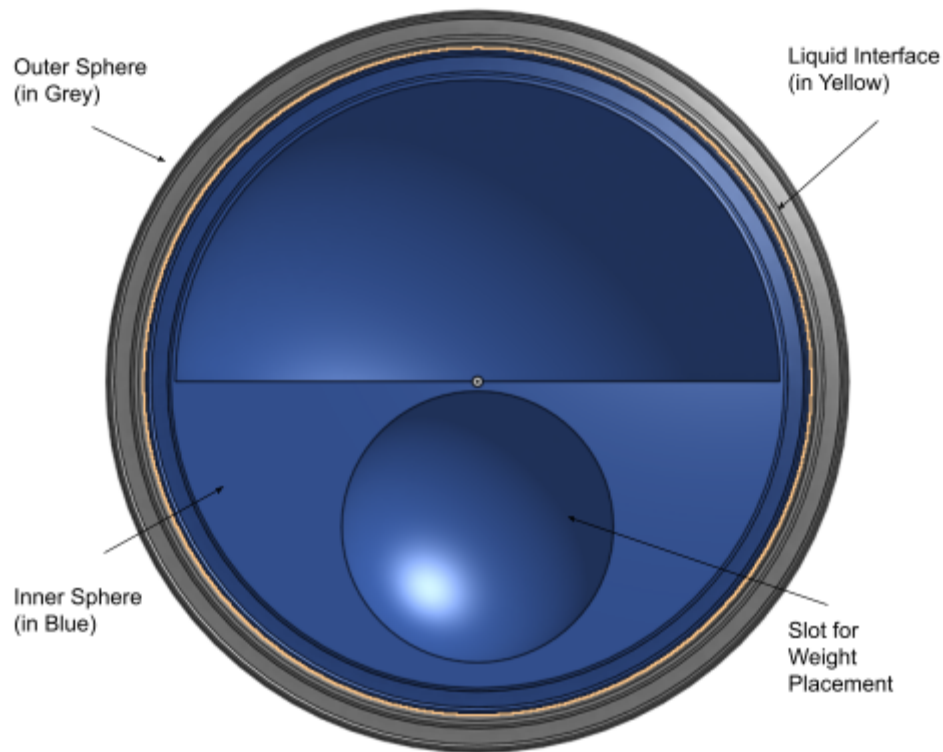


Fig. 8. Labeled diagram of assembled Kaliedosphere (half-view).

Operation of the Final Design

The operation of the toy is straightforward, engaging, and intuitive for users. Below is the sequence of steps involved:

1. Initial Interaction:

- The user picks up the toy and observes its spherical shape and transparency. The moiré patterns on the inner and outer spheres are immediately visible.

2. Rotation or Rolling:

- The user rotates or rolls the toy in their hands or on a flat surface. The smooth water interface allows the inner sphere to maintain its upright orientation, while the outer sphere moves independently.

3. Moiré Pattern Dynamics:

- As the toy rotates, the relative motion between the hydro-dipped patterns on the two spheres generates dynamic moiré effects. The patterns appear to move,

distort, or shift, captivating the user's attention and enhancing their spatial reasoning and visual perception.

4. **Repeat and Explore:**

- The user continues to interact with the toy, exploring different speeds and angles of rotation to create varied visual effects.

Validation

The final design effectively satisfies the user needs and engineering requirements established earlier in the report, as demonstrated through design features, material choices, and user testing. The use of hydro-dipping ensures that the moiré patterns on the inner and outer spheres are intricate and high-quality, achieving the visual stimulation critical for engaging children. This feature was validated informally through user testing at the Senior Design showcase, where Brandon Stafford's son interacted with the toy and found it highly engaging. His reaction confirmed that the design successfully meets its educational and entertainment goals by captivating the user's attention and enhancing spatial reasoning.

The water interface between the inner and outer spheres minimizes friction, enabling the inner sphere to maintain smooth and consistent movement relative to the outer sphere. This design aspect ensures seamless operation of the toy during use. Additionally, the choice of polycarbonate for both the inner and outer spheres offers the necessary transparency to highlight the moiré patterns while providing the durability required for frequent handling and accidental drops, addressing the need for safety and robustness.

The intuitive nature of the design allows children to quickly understand and engage with the toy without the need for instructions, enhancing its appeal as both an educational and recreational product. By rolling or rotating the sphere, users experience dynamic moiré effects that maintain their interest and curiosity. The innovative combination of these features results in a product that is not only visually and functionally compelling but also manufacturable at scale, aligning with the identified user needs of affordability, durability, safety, and marketability.

Production Plan and Social/Ethical Considerations

Production Plan

To create a manufacturable final product, the design and fabrication methods must prioritize scalability, cost-efficiency, and quality. The primary production method ideally used for the Kaleidosphere toy is injection molding, which is highly suitable for producing spherical components at scale. Based on our technical drawings (see Appendix IV), the following steps outline the proposed production process:

1. Material Selection and Sourcing

Polycarbonate has been chosen for its durability, transparency, and suitability for hydro-dipping. Materials will be sourced from certified suppliers to ensure consistency and compliance with safety standards.

2. Injection Molding

Custom molds designed in CAD will be used to produce both the inner and outer spheres. The injection molding process will ensure precise tolerances and consistent dimensions, with a targeted fabrication time of less than 30 minutes per unit.

3. Hydro-Dipping for Moiré Patterns

Hydro-dipping will be employed to apply high-quality, visually engaging moiré patterns to the inner and outer spheres. This technique ensures uniformity and durability of the patterns, enhancing the visual appeal of the toy.

4. Assembly

The inner sphere will be weighted, using the circular slot for placement, to maintain its orientation and securely sealed to ensure durability and prevent leakage. The outer sphere will be joined around the inner sphere using a press-fit design, further secured with adhesive to prevent separation during play.

5. Quality Assurance

Each unit should undergo visual inspection and durability testing to ensure that it meets the engineering requirements for safety, functionality, and visual effect.

By integrating these steps, the production plan ensures that the Kaleidosphere is manufacturable at scale while maintaining its affordability, safety, and appeal.

Social and Ethical Concerns

While the Kaleidosphere is designed to be a safe and engaging toy, several potential social and ethical issues should be considered:

1. Environmental Impact

The use of polycarbonate and hydro-dipping materials raises concerns about the environmental footprint of our product. To mitigate this, we recommend exploring recyclable or biodegradable materials for future iterations. Additionally, minimizing material waste during production will reduce the toy's overall environmental impact.

2. Economic Accessibility

Although affordability is a key design goal, ensuring that the toy remains accessible to a diverse audience—including underfunded schools and lower-income families—may require partnerships with educational institutions or nonprofit organizations to subsidize costs.

3. Safety Concerns

Ensuring compliance with ASTM F963 standards addresses potential safety concerns. However, unintended misuse, such as dismantling by curious children, could lead to hazards. Clear labeling and robust design choices will minimize these risks.

By proactively addressing these social and ethical considerations, the Kaleidosphere aims to maximize its educational and entertainment value while minimizing potential negative impacts.

Conclusions and Future Work

Strengths and Weaknesses of Current Design

The design exhibits several notable strengths that align well with user needs and engineering requirements. Foremost among these is the visual engagement offered by the dynamic moiré patterns. The hydro-dipping technique ensures intricate and high-quality designs that captivate users, fulfilling the goal of creating a visually stimulating toy. Additionally, the water interface provides a frictionless medium that enhances the smooth rotational movement of the inner sphere, contributing to seamless operation and preserving the integrity of the patterns over time.

The choice of polycarbonate material is another strength, offering exceptional clarity to display the moiré effects while ensuring the durability necessary for frequent use. This material is also impact-resistant, addressing the safety and robustness needs critical for a child-focused product. Moreover, the intuitive operation of the toy allows children to engage with it immediately, making it both educational and entertaining without requiring instructions. The scalable manufacturability achieved through injection molding and hydro-dipping ensures that the toy can be produced efficiently and cost-effectively, supporting marketability and accessibility.

However, the design also has areas for improvement. While the hydro-dipping process produces high-quality patterns, it may introduce variability in pattern alignment during production, potentially affecting the consistency of the moiré effect. The use of water as a friction-reducing medium introduces a potential long-term maintenance challenge, as any leakage or evaporation could compromise the toy's functionality. Additionally, user testing has been limited to informal interactions, such as feedback from Brandon Stafford's son at the Senior Design showcase. More comprehensive and systematic user testing, including diverse age groups and controlled evaluations, would provide stronger validation of the toy's performance and engagement.

Another limitation involves the reliance on polycarbonate material for the spheres. While polycarbonate is ideal for strength and clarity, its higher production costs compared to alternatives may impact affordability, especially in large-scale manufacturing. Exploring cost-effective materials without compromising performance could address this concern.

Recommendations for Future Revisions

While the design of the Kaleidosphere demonstrates the potential for creating a visually engaging and educational toy, challenges in the fabrication process highlight areas for improvement. The injection molding process, in particular, did not yield successful parts during this project, emphasizing the need for future refinements. Below are the primary recommendations for revisions to the design solution:

1. Implement Computational Fluid Dynamics (CFD) Analysis

- **Rationale:** The failure of the injection molding process was largely attributed to the irregularities of polymer flow and possible uneven cooling during the molding process. Conducting a CFD analysis would provide critical insights into these issues, enabling optimization of the mold design and manufacturing parameters.
- **Proposed Action:** Utilize software such as COMSOL or ANSYS to simulate material flow with dynamic viscoelasticity and cooling dynamics. These simulations would help address issues related to material distribution, gate placement, and cooling rates, improving the success rate of molded parts.

2. Access Advanced Molding Equipment

- **Rationale:** The available equipment lacked the capability to reach the high temperatures required for molding polycarbonate, a critical material for the design. Upgrading to industrial-grade molding equipment would be essential for producing high-quality parts.
- **Proposed Action:** Collaborate with facilities that have advanced injection molding capabilities or explore external manufacturing partners with expertise in polycarbonate molding.

3. Redesign Molding Approach

- **Rationale:** The spherical geometry of the toy presents unique challenges for material flow during molding. Current molds and single-point injection methods may not adequately address these challenges.
- **Proposed Action:** Redesign the molds to incorporate multi-gate or rotational molding techniques, ensuring more uniform material flow and reducing defects.

Experimentation with alternative molding directions could also improve material distribution and part integrity.

4. Enhance Hydro-Dipping Precision

- **Rationale:** While hydro-dipping produces visually compelling patterns, achieving consistent alignment remains a challenge. Misaligned patterns can detract from the overall quality of the moiré effect.
- **Proposed Action:** Develop custom fixtures or guides to ensure precise alignment of the patterns during hydro-dipping. Automating the process may further enhance repeatability and scalability.

AI Attribution

AI tools were used to assist in the creation of this report. Specifically, ChatGPT was utilized for various tasks, including refining and editing sections such as the problem statement, motivation, and engineering requirements. Additionally, it was used to find relevant sources to research in order to support arguments in the motivation section and to draft some of the sections following the competitor analysis, such as concept generation and selection. All content generated with AI was thoroughly reviewed and edited by the team to ensure accuracy, coherence, and alignment with project goals.

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Appendices – Technical Documentation

Appendix I: Concepts

Motion applied to the outer sphere by the user	The user can see the Moire effect working	The user can orient the outer sphere independently of the inner sphere	The user should only interact with the outer sphere	The user can see the pattern on the inner sphere
[Rolled across floor]	[“Trippy” patterns]	[Doesn’t matter - inner sphere not fixed]	[Outer sphere sealed with glue]	[Sand and Polish]
[Contact juggling - like a Fushigi ball]	[Image moving - i.e. horse running]	[Inner sphere fixed with weight]	[Outer sphere sealed with acrylic sealant - dichloromethane]	[Use existing clear hemispherical parts]
[Mechanical initiation - e.g. pull level]	[Two transparent layers with printed moire patterns]	[Inner sphere fixed with magnet]		[Illuminate internally]
[Electrical initiation - e.g. push button]	[Projected or reflective patterns inside of spheres]	[Gyroscopic internal mechanism]		

Fig. 6. Morphology of Kaleidosphere toy.

Prototype I: Christmas-Ornament Design, Taped Patterns

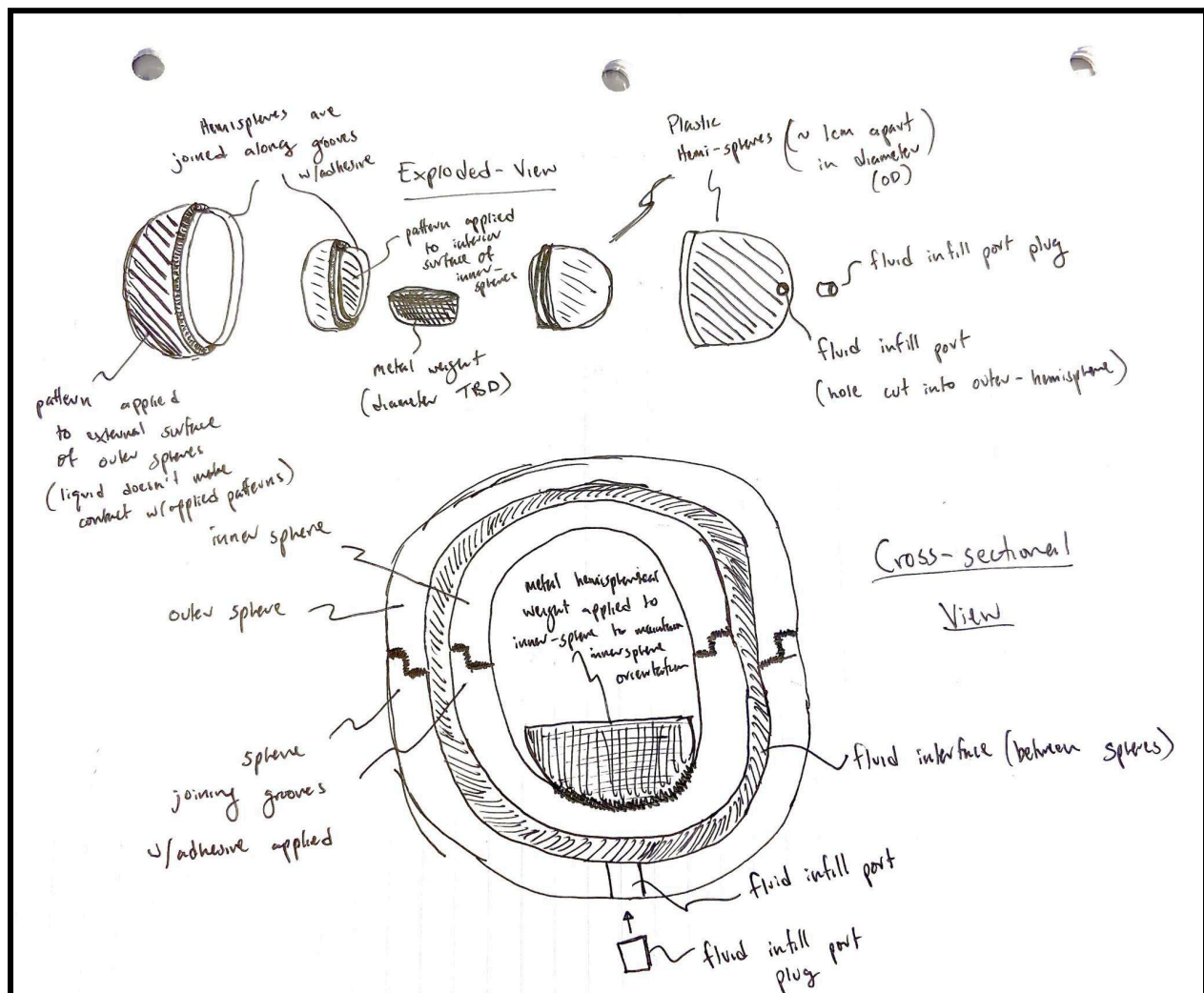


Fig. 5. Initial Concept Design Sketch.

Prototype II: Injection Molded Design, Hydro-dipped Patterns

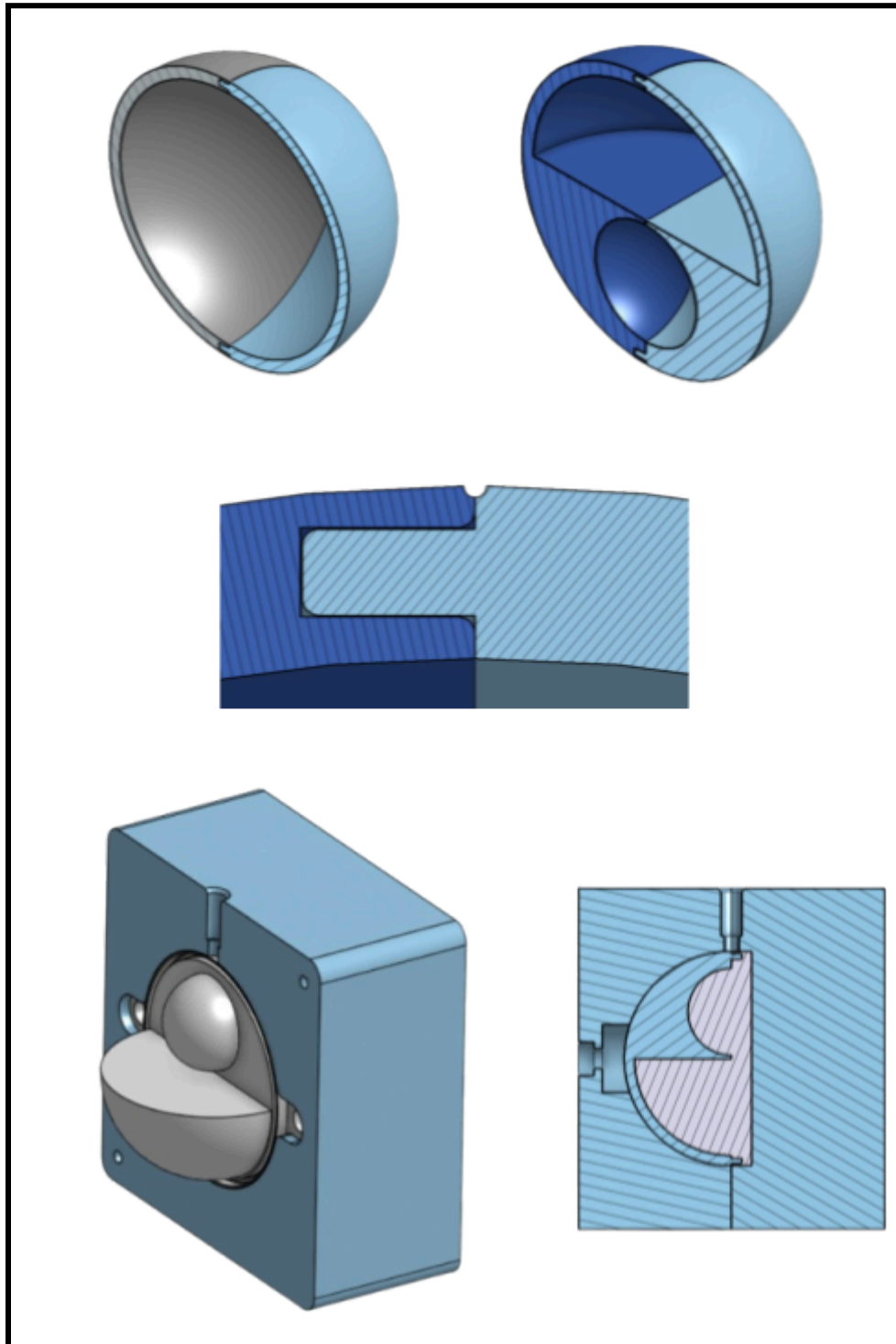


Fig. 3. Injection molded parts concept and mold design.

Appendix II: Matrix of user needs and engineering requirements

User Needs → Eng. Reqs. ↓	Visually Stimulating and Engaging	Affordable	Safe for Repeated Play	Durable	Lightweight	Unique Use of Moiré Patterns	Marketable	Manufacturable at Scale
Effective Moiré Effect	X					X		
Weight			X		X			
Manufacturing Cost		X						X
Dimensions			X					
Durability				X				
Manufacturing Efficiency								X
Operational Lifetime							X	
Interactive Moiré Patterns	X					X		
Material Safety and Compatibility			X					

Fig. X. Matrix of user needs (horizontal) and engineering requirements (vertical).

Appendix III: Bill of materials

Bill of Materials						
Product: Kaleidospheres				Date: Dec 18, 2024		
Assembly: Kaleidosphere						
Item #	Part #	Qty	Name	Material	Source	Cost (per unit)
1	ASY-IS-01	1	Inner Sphere	Polycarbonate	Custom Part	\$4.53
2	TWN-WTR	1	Liquid Interface	Water	Town Water	\$2.72
3	ASY-OS-01	1	Outer Sphere	Polycarbonate	Custom Part	\$0.37
4	ASY-HD-01	1	Hydro-dipping Materials	Hydrographics Film	Liquid Concepts	\$2.28
Team Member: Ethan Goldman			Prepared by: Ethan Goldman			
Team Member: Tyler Pisinski			Checked by: Tyler Pisinski			
Team Member: May Wildgrube			Approved by: May Wildgrube			

Bill of Materials						
Product: Kaleidospheres				Date: Dec 18, 2024		
Assembly: Inner Sphere (Prt# ASY-IS-01)						
Item #	Part #	Qty	Name	Material	Source	Cost (per unit)
1	PRT-IHS1-01	1	Hemi-Sphere I	Polycarbonate	Custom Part	\$0.13
2	PRT-IHS2-01	1	Hemi-Sphere II	Polycarbonate	Custom Part	\$0.12
3	B002665SYY	1	White Paint	Acrylic Paint	Amazon	\$1.75
4	B0CM9DCX6 W	1	Ball Bearings Weight	Steel	Amazon	\$0.02
5	B0149IG548	1	Adhesive	Dichloromethane	Amazon	\$0.23
6	ASY-HD-01	1	Hydro-dipping Materials	Hydrographics Film	Liquid Concepts	\$2.28
Team Member: Ethan Goldman			Prepared by: Ethan Goldman			
Team Member: Tyler Pisinski			Checked by: Tyler Pisinski			
Team Member: May Wildgrube			Approved by: May Wildgrube			

Bill of Materials						
Product: Kaleidospheres				Date: Dec 18, 2024		
Assembly: Outer Sphere (Prt# ASY-OS-01)						
Item #	Part #	Qty	Name	Material	Source	Cost (per unit)
1	PRT-OHS1-01	1	Hemi-Sphere I	Polycarbonate	Custom Part	\$0.07
2	PRT-OHS2-01	1	Hemi-Sphere II	Polycarbonate	Custom Part	\$0.07
3	B0149IG548	1	Adhesive	Dichloromethane	Amazon	\$0.23
Team Member: Ethan Goldman			Prepared by: Ethan Goldman			
Team Member: Tyler Pisinski			Checked by: Tyler Pisinski			
Team Member: May Wildgrube			Approved by: May Wildgrube			

Bill of Materials						
Product: Kaleidospheres				Date: Dec 18, 2024		
Assembly: Hydro-dipping Materials (Prt# ASY-HD-01)						
Item #	Part #	Qty	Name	Material	Source	Cost (per unit)
1	B07VSWK8TF	1	Hydrographics Film	Hydrographics Film	Amazon	\$1.20
2	ACTIVATOR-AEROSOL	1	Hydrographics Activator	Hydrographics Activator	Liquid Concepts	\$0.40
3	AP-01	1	Adhesion Promoter	Adhesion Promoter	Liquid Concepts	\$0.28
4	AERO-1K-GL OSS-C100	1	1K Gloss Clear Coat Aerosol	1K Gloss Clear Coat Aerosol	Liquid Concepts	\$0.40
Team Member: Ethan Goldman			Prepared by: Ethan Goldman			
Team Member: Tyler Pisinski			Checked by: Tyler Pisinski			
Team Member: May Wildgrube			Approved by: May Wildgrube			

Appendix IV: Technical drawings

See attached PDFs on following pages. Attached PDFs include:

Mold Assy Concept (Mold shown is for PRT-OHS1-01)

PRT-IHS1-01

PRT-IHS2-01

PRT-OHS1-01

PRT-OHS2-01

Kaleidosphere Full Assembly

Appendix V: Software code, if applicable

Not Applicable

Appendix VI: Video

Please find the video attached here:

 Senior Design Kaleidosphere Final Report Video - Made with Clipchamp_173457004752...